

Pharmaceutical Sales Forecasting Using GM(1,1) and SVR

1. Business Problem

The pharmaceutical industry is highly dynamic, with demand for medicines and healthcare products influenced by seasonal factors, market campaigns, and consumer behavior. Accurate sales forecasting is crucial for:

- Inventory management: Avoid stockouts or overstocking
- Production planning: Align manufacturing schedules with demand
- Marketing strategy: Optimize promotional campaigns
- Financial planning: Predict revenue and manage cash flow

Challenges:

- Volatile demand: Sales fluctuate due to seasonal illness patterns.
- Data sparsity: Historical sales data may contain missing values or outliers.
- Time-dependent trends: Past sales trends and seasonality impact future sales.
- Non-linear patterns: Simple linear models fail to capture complex dependencies.

2. Objective

To build a robust sales forecasting system for pharmaceutical products using:

- GM(1,1) model (Grey Model): Suitable for small datasets and captures overall trends.
- Support Vector Regression (SVR): Captures non-linear patterns and provides accurate short-term forecasts.

Goal:

Predict future sales accurately to improve inventory planning, marketing, and financial decision-making.

3. Data Description

The dataset includes historical e-commerce sales of pharmaceutical products.

Column Name	Description
Date	Date of transaction
Product_ID	Unique identifier for the product
Product_Name	Name of the product
Category	Product category
Units_Sold	Number of units sold (used as Sales_Value)
Price_per_Unit (Rs.)	Unit price
Revenue (Rs.)	Total revenue
Discount (%)	Discount applied
Customer_Rating	Average rating given by customers
Stock_Availability (%)	Stock availability in percentage
Marketing_Spend (Rs.)	Marketing expenses
Season	Season information (e.g., winter, monsoon)
Forecast_Target	Target variable for next-day sales forecasting

Pharmaceutical Sales Forecasting Using GM(1,1) and SVR

Data Preprocessing Steps:

- Handle missing values using linear interpolation and rolling mean.
- Remove or cap outliers using IQR method.
- Generate additional features: month, quarter, seasonality (sin/cos), lag features.

4. Modeling Approach

4.1 GM(1,1) - Grey Forecasting Model

Suitable for small or incomplete datasets. Captures overall trend in sales over time.

- Convert original series into cumulative series.
- Compute mean sequence and parameters using least squares.
- Generate forecasted sales series.

4.2 SVR - Support Vector Regression

Handles non-linear relationships in time series.

- Scale features (TimeIndex) and target (Sales_Value) using StandardScaler.
- Train SVR with RBF kernel.
- Forecast both in-sample and future 6 months.

Benefits: Robust to outliers; Captures complex sales patterns.

5. Visualization

Actual vs Forecasted Sales: Compare GM(1,1) and SVR predictions with actual sales.

Future Sales Forecast: SVR predicts the next 6 months to guide production and inventory planning.

Graph Features:

- X-axis: Date
- Y-axis: Sales Value
- Lines: Actual Sales, GM(1,1) Forecast, SVR In-Sample, SVR Future Forecast

6. Business Impact

Implementing this forecasting system enables the pharmaceutical business to:

- Maintain optimal inventory levels.
- Plan marketing campaigns more effectively based on forecasted demand.
- Reduce stockouts and overstocks, saving costs.
- Make data-driven decisions for production and financial planning.

7. Future Enhancements

- Incorporate multi-variate SVR using features like marketing spend, seasonality, and discounts.
- Use advanced deep learning models (LSTM, Transformer) for long-term forecasting.
- Deploy a real-time dashboard for sales predictions.